

Day 1 Tasks

Task 1: Problem Definition and Success Metric

The goal was to predict whether a person earns more than 50K per year, which stands in for a real business case such as targeted marketing or credit screening. The positive class is income above 50K. Because the classes are imbalanced, accuracy alone is misleading, so precision was chosen as the primary metric. Every predicted positive triggers an outreach cost, so a wrong guess is more expensive than a missed opportunity. Recall, F1, and ROC AUC were tracked as supporting metrics.

Task 2: Data Load and Quick EDA

The Adult dataset was loaded and the target was converted to a binary 0 or 1 value. Missing values, marked as question marks in the raw data, were converted to proper nulls. Basic exploration covered the shape and types of the data, numeric summaries, value counts for categorical columns, and simple charts to see how features like education and hours worked relate to income.

Task 3: Reproducible Splits

The data was split into a training set, a small development set for quick checks, and a hold out test set of about 20 percent, stratified so the class balance stays consistent across all three. The hold out set was kept completely separate from any tuning, so the final numbers reflect how a model would likely perform on new, unseen people rather than data it has already learned from.

Task 4: Simple Baselines

Two baselines were built and compared on the hold out set. A majority class predictor that always guesses the more common outcome, and a rule based classifier that predicts a high income if education level is high or capital gain is present. The majority class baseline scored well on accuracy but had no real predictive value, since it never identifies a high earner. The rule based baseline performed better on every metric that matters for this goal, showing that even one or two simple features carry real signal.

Task 5: Error Analysis and Next Steps

A sample of false positives and false negatives was reviewed to look for patterns. False positives tended to have high education but lower paying roles or fewer working hours, meaning education alone overstates income. False negatives often had no capital gain and worked long hours in roles the simple rule does not account for. The main next steps are:

- Properly encode categorical features instead of ignoring them
- Fill in or flag missing values in workclass, occupation, and native country
- Handle the heavy skew in capital gain and capital loss
- Group rare categories in high cardinality columns
- Remove duplicate or overlapping features
- Explore feature interactions, such as hours worked combined with occupation

Precision remains the primary metric going forward, since the baseline results already show it as the biggest gap between a usable model and one that simply looks accurate on paper.